STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 11: Thermodynamics

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculations and units. Original practice material, not an official examination.

State the zeroth law. [1]
Define internal energy. [1]
First law with W done by system? [1]
Meaning of adiabatic? [1]
Meaning of isothermal? [1]
Can a cyclic heat engine be 100% efficient? [1]
Ideal gas equation? [1]
Carnot efficiency? [1]

At constant volume, 500 J heat added: work? [2]
Gas absorbs 800 J and does 300 J work: DeltaU? [2]
Carnot T_h=600K,T_c=300K: efficiency? [2]
Ideal-gas isothermal process: DeltaU? [2]
1 mol with C_v=20 J/mol K and DeltaT=5 K: DeltaU? [2]
Isothermal versus adiabatic? [2]

Gas absorbs 1000 J and does 400 J work: DeltaU? [3]
Carnot T_h=500K,T_c=300K: ? [3]
Isobaric gas expands .01 to .03 m^3 at 100 kPa. Work? [3]
Why no real cyclic engine converts all heat to work? [3]

Explain reversible and irreversible processes with examples. [4]
Engine absorbs 1500 J from 600K, rejects to 300K; actual work=500J. Find Carnot limit, actual efficiency and Qout. [4]

End of paper - check working, units, signs and significant figures.